

Matt Bowring

mbowring@purdue.edu | Cambridge, MA

[Website](#) | [GitHub](#) | [LinkedIn](#)

Experience

Taconic Trading LLC

Software Engineer

C++, Python, CMake, Git

Jan. 2026 - Present

New York City, NY (remote)

- Develop infrastructure for real-time financial data to enable low-latency automated trading

The MathWorks

Software Engineer (Math/PDE)

MATLAB, C++, Python, CMake, OpenMP, Git

May 2022 - Present

Natick, MA (on-site)

- Lead development of the [MATLAB Quantum Computing Library](#) (unreleased prototype to 9,300+ users)
 - Design object-oriented MATLAB framework to enable quantum programming across Windows, Linux, and Mac
 - Develop templated C++ state-vector simulator for CPU/GPU with OpenMP multi-threading
 - Develop graph-based C++ compiler to generate optimized assembly IR code targeting hardware backends
 - Build client-side REST infrastructure using APIs from AWS and IBM to submit batched jobs to hardware
 - Implement sparse algorithms for estimation of Pauli observables, error mitigation, and gate decomposition
 - Manage CI/CD, write all unit/functional/performance tests, and analyze benchmarks against other quantum libraries
- Consulted industry users on machine learning and quantum computing applications
 - Implemented quantum neural networks for image classification ([SPIE publication](#)) and reinforcement learning
 - Researched quantum optimization methods using Pauli correlation encoding, quadratization, and penalty dephasing
 - Developed presentations and examples for industry conferences (ACS 25, AWS re: Invent 25, IBM QDC 22-25)

The MathWorks

Software Engineer

MATLAB, C++, Python, Git

May 2021 - May 2022

Natick, MA (on-site)

- Trained GNN on QM7-X dataset to predict equilibrium energy using custom pipeline to encode molecular properties
- Developed REST interface to cloud-based quantum annealing hardware and benchmarked minor-embedding heuristics
- Wrote parallel simulations to compute qubit transition probabilities for truncated graph Hamiltonian across coupling strength and energy separation parameters

Education

Purdue University (3.8 GPA)

M.S. Electrical/Computer Engineering

Jan. 2025 - Jan. 2028

West Lafayette, IN (remote)

- Fabricated mixed-signal PCB to solve combinatorial problems using phase dynamics of LC oscillators ([arXiv preprint](#))
 - Developed simulations of coupled oscillators in LTspice and modeled noisy phase dynamics with MATLAB
 - Designed schematics and layouts for two multi-layer boards in Altium Designer and soldered components
 - Developed real-time embedded C++ firmware using SPI and FreeRTOS to measure phases at 1MHz
 - Automated test instruments with MATLAB to control perturbations and analyze Fourier spectrum

The University of New Hampshire (3.9 GPA)

B.S. Mechanical Engineering

Aug. 2017 - May 2021

Durham, NH (on-site)

- Awarded over \$100,000 in merit-based academic scholarships for developing an autonomous quadcopter
- Led the Quadcopter Engineering Team; Volunteered as Makerspace Administrator, Academic Tutor and Mentor

Projects

Computer Systems

2024 - Present

- Operate custom multi-GPU Linux machine to accelerate research experiments; Familiar with CPU/GPU/FPGA architectures, kernels, and SIMD parallelism; Broad interest in programming language semantics

Machine Learning

2020 - Present

PyTorch/CUDA, NetworkX, OpenCV

- Implemented greedy algorithms to schedule and place tensors on mock dataflow architecture to minimize data movement
- Reproduced edge-of-chaos Lyapunov analysis in randomly initialized Transformers to explore trainability and heavy-tailed self-regularization
- Implemented Langevin MCMC with skew-symmetric drift and heavy-tailed noise to study accelerated convergence in combinatorial optimization.
- Trained CNN/RNN on driving footage to predict dash-camera frame misalignment by tracking vanishing points